

Advanced Microeconomics

Dr. Juliane Hennecke

Nils Sulzmann

15.12.2025, Winter term 2025/26

Problem Set 11: General equilibrium and Edgeworth box

1) Consider a pure-exchange, private-ownership economy, consisting in two consumers, denoted by $i = 1, 2$, who trade two commodities, denoted by $l = 1, 2$. Each consumer i is characterized by an endowment vector, $\omega_i \in \mathbb{R}_+^2$, a consumption set, $X_i = \mathbb{R}_+^2$, and regular and continuous preferences, $\succsim_i$ on X_i .

a) Assuming that the consumers' endowments are $\omega_1 = (1, 2)$ and $\omega_2 = (2, 1)$, respectively, construct the Edgeworth Box relative to economy under consideration. With reference to the same economy, define the following notions: competitive equilibrium, Pareto-efficient allocation, Pareto set, contract curve.

b) Find the equation describing the Pareto set (internal solutions); then, taking commodity 1 as the numeraire, hence positing $p_1 = 1$, find the competitive equilibrium allocation and price system; and, finally, draw your results in the Edgeworth Box in each of the following two cases:

b.1 both consumers' preferences are represented by the same Cobb-Douglas utility function:

$$u_i(x_{1i}, x_{2i}) = x_{1i}^{1/3} x_{2i}^{2/3}$$

b.2 the two consumers' preferences are respectively represented by the following quasilinear utility functions: $u_1(x_{11}, x_{21}) = x_{11} + \ln x_{21}$; $u_2(x_{12}, x_{22}) = x_{12} + 2 \ln x_{22}$

c) Explain in which of the above two cases the preferences are homothetic and in which case, instead, the preferences are such as to rule out the wealth effects usually associated with price changes. How such peculiar properties of consumers' preferences affect the Pareto set and the shadow prices, i.e., the prices implicit in the Pareto-efficient allocations?